Solving Absolute Value Inequalities

The "In or Out" Method — Step-by-Step Practice

Name: _____ Date: _____ Period: _____

The Big Rule

Absolute value bars ask: how far is the inside expression from 0? Instead of memorizing which symbol means *and* and which means *or*, **test the middle point** (where the inside expression equals 0) to see whether it belongs to the solution:

Test result at the middle	Region	Compound inequality
True	**in** (between the boundaries)	**and**
False	**out** (outside the boundaries)	**or**

Every practice problem below uses the **same three steps**: (1) find the boundaries, (2) test the middle point True/False, (3) write *and*/*or* and solve. Fill in every blank!

Example 1 Solve $|3x - 4| \leq 8$ (an "And" case)

Step 1 Find the boundaries

Since $|-8| = 8$ and $|8| = 8$, the expression $3x - 4$ hits the edge at -8 and 8. Mark both as **closed** dots (the inequality is $\leq$, so the ends are included).



Step 2 Check: In or Out?

Test the middle point $3x - 4 = 0$:

$$|0| = 0 \leq 8 \quad \textbf{True}.$$

The point *between* the boundaries works, so we keep the **in** region. That makes this an **and** statement (a *between* inequality), not an *or*.

Step 3 Write the compound inequality and solve

Because it is "in," $3x - 4$ stays *between* -8 and 8:

$$-8 \leq 3x - 4 \leq 8.$$

Solve all three parts at once. Add 4:

$$-8+4 \le 3x-4+4 \le 8+4 \qquad \Longrightarrow \qquad -4 \le 3x \le 12.$$

Divide by 3:

$$\frac{-4}{3} \le x \le \frac{12}{3} \qquad \Longrightarrow \qquad -\frac{4}{3} \le x \le 4.$$

Solution: $-\frac{4}{3} \le x \le 4$ or $\left[-\frac{4}{3}, 4\right]$



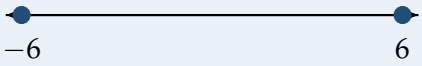
Check: at $x = 0$, $|3(0)-4| = 4 \le 8$ ✓

Example 2 Solve $|2x+4| \ge 6$ (an "Or" case)

Same method as Example 1 – but this time the test point will fail, which flips it to an *or* statement.

Step 1 Find the boundaries

Since $|-6| = 6$ and $|6| = 6$, the expression $2x+4$ hits the edge at -6 and 6. Mark both as **closed** dots (the inequality is $\ge$, so the ends are included).



Step 2 Check: In or Out?

Test the middle point $2x+4=0$:

$$|0| = 0 \ge 6 \quad \textbf{False}.$$

The point *between* the boundaries does **not** work, so the solution is the **out** region. That makes this an **or** statement (an *outside* inequality), not an *and*.

Step 3 Write the compound inequality and solve

Because it is "out," $2x+4$ stays *outside* -6 and 6 — less than or equal to the left boundary, **or** greater than or equal to the right boundary:

$$2x+4 \le -6 \qquad \textbf{or} \qquad 2x+4 \ge 6.$$

Solve each piece on its own. Subtract 4 from both sides:

$$2x \le -10 \qquad \textbf{or} \qquad 2x \ge 2.$$

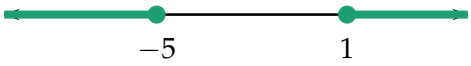
Divide both sides by 2:

$$x \le -5 \qquad \textbf{or} \qquad x \ge 1.$$

Common Mistake to Avoid

The original inequality is $\geq$, not a strict $>$ – so the boundary points themselves are part of the solution, and the compound inequality must use $\leq$ and $\geq$, not $<$ and $>$. Writing "$2x+4<-6$ or $2x+4>6$" would quietly kick $x=-5$ and $x=1$ back out of the answer. Always check: plug the boundary value into the *original* inequality — at $x=-5$, $|2(-5)+4|=|-6|=6\geq 6$ ✓, so it belongs, and the dots on the number line stay **closed**.

Solution: $x\leq -5$ or $x\geq 1$ or $(-\infty,-5]\cup[1,\infty)$



Check: at $x=-10$ (outside), $|2(-10)+4|=16\geq 6$ ✓ at $x=0$ (between), $|2(0)+4|=4\geq 6$? ✗ (correctly excluded)

Key Idea — And vs. Or at a Glance

Form	Test at the middle	Solution shape
$\lvert\text{expr}\rvert\leq k$	always True (In)	**and**: one connected interval
$\lvert\text{expr}\rvert\geq k$	always False (Out)	**or**: two separate rays

(Assuming $k>0$.) The middle-point test is a shortcut for the graph of $y=|\text{expr}|$: it dips *below* the line $y=k$ only in the middle (giving an *and*), so anywhere it stays at or above $y=k$ gives an *or*.

Part A Level 1 — Direct Absolute Value $|x| \square c$

Follow the three steps. The boundaries of $|x|$ are just $x = c$ and $x = -c$.

1. $|x| \leq 4$

Step 1 — Boundaries: $x =$ _____ and $x =$ _____

Step 2 — Test $x = 0$: $|0|$ ____ $4 \Rightarrow$ True / False (circle one)

In / Out $\Rightarrow$ and / or (circle one)

Step 3: _____ $\leq x \leq$ _____

Answer: _____

Step 4 — Graph it:

2. $|x| \geq 3$

Step 1 — Boundaries: $x =$ _____ and $x =$ _____

Step 2 — Test $x = 0$: $|0|$ ____ $3 \Rightarrow$ True / False (circle one)

In / Out $\Rightarrow$ and / or (circle one)

Step 3: $x \leq$ _____ or $x \geq$ _____

Answer: _____

Step 4 — Graph it:

3. $|x| < 6$

Step 1 — Boundaries: $x =$ _____ and $x =$ _____

Step 2 — Test $x = 0$: $|0|$ ____ $6 \Rightarrow$ True / False (circle one)

In / Out $\Rightarrow$ and / or (circle one)

Step 3: _____ $< x <$ _____

Answer: _____

Step 4 — Graph it:

4. $|x| > 2$

Step 1 — Boundaries: $x =$ _____ and $x =$ _____

Step 2 — Test $x = 0$: $|0|$ ____ $2 \Rightarrow$ True / False (circle one)

In / Out $\Rightarrow$ and / or (circle one)

Step 3: $x <$ _____ or $x >$ _____

Answer: _____

Step 4 — Graph it:

5. $|x| \leq 0$

(Hint: absolute value can never be negative — when can it equal exactly 0?)

Answer: _____

Graph it:

6. $|x| \geq 0$

(Hint: absolute value is always ≥ 0 — for which x is this ever false?)

Answer: _____

Graph it:

Part B Level 2 — Shift Inside $|x \pm a| \square c$

Same three steps — but the middle point is no longer $x = 0$; solve $x \pm a = 0$ first.

1. $|x - 2| \leq 5$

Step 1: $x - 2 =$ _____ and $x - 2 =$ _____ $\Rightarrow x =$ _____ and $x =$ _____

Step 2 — Test middle ($x = 2$): $|0|$ ___ 5 $\Rightarrow$ True/False

In/Out $\Rightarrow$ and/or (circle)

Step 3: _____$\leq x \leq$ _____

Answer: _____

Step 4 — Graph it:

2. $|x + 4| \geq 3$

Step 1: $x + 4 =$ _____ and $x + 4 =$ _____ $\Rightarrow x =$ _____ and $x =$ _____

Step 2 — Test middle ($x = -4$): $|0|$ ___ 3 $\Rightarrow$ True/False

In/Out $\Rightarrow$ and/or (circle)

Step 3: $x \leq$ _____ or $x \geq$ _____

Answer: _____

Step 4 — Graph it:

3. $|x - 5| < 2$

Step 1: $x - 5 =$ _____ and $x - 5 =$ _____ $\Rightarrow x =$ _____ and $x =$ _____

Step 2 — Test middle ($x = 5$): $|0|$ ___ 2 $\Rightarrow$ True/False

In/Out $\Rightarrow$ and/or (circle)

Step 3: _____$< x <$ _____

Answer: _____

Step 4 — Graph it:

4. $|x + 1| > 4$

Step 1: $x + 1 =$ _____ and $x + 1 =$ _____ $\Rightarrow x =$ _____ and $x =$ _____

Step 2 — Test middle ($x = -1$): $|0|$ ___ 4 $\Rightarrow$ True/False

In/Out $\Rightarrow$ and/or (circle)

Step 3: $x <$ _____ or $x >$ _____

Answer: _____

Step 4 — Graph it:

5. $|x + 3| \leq 6$

Step 1: $x + 3 =$ _____ and $x + 3 =$ _____ $\Rightarrow x =$ _____ and $x =$ _____

Step 2 — Test middle ($x = -3$): $|0|$ ___ 6 $\Rightarrow$ True/False

In/Out $\Rightarrow$ and/or (circle)

Step 3: _____$\leq x \leq$ _____

Answer: _____

Step 4 — Graph it:

6. $|x - 7| \geq 5$

Step 1: $x - 7 =$ _____ and $x - 7 =$ _____ $\Rightarrow x =$ _____ and $x =$ _____

Step 2 — Test middle ($x = 7$): $|0|$ ___ 5 $\Rightarrow$ True/False

In/Out $\Rightarrow$ and/or (circle)

Step 3: $x \leq$ _____ or $x \geq$ _____

Answer: _____

Step 4 — Graph it:

Part C Level 3 — Isolate First! Get $|\cdot|$ alone, then run the 3 steps

Add, subtract, or divide to isolate the absolute value first — just like Example 2 required solving for $2x+4$ before testing.

1. $|x|+3 \le 7$

Isolate: $|x| \le$ _____

Step 1 — Boundaries: $x =$ _____ and $x =$ _____

Step 2 — Test $x = 0$: True / False

In / Out ⇒ and / or (circle)

Step 3: _____$\le x \le$ _____

Answer: _____

Graph it:

2. $2|x| \ge 10$

Isolate: $|x| \ge$ _____

Step 1 — Boundaries: $x =$ _____ and $x =$ _____

Step 2 — Test $x = 0$: True / False

In / Out ⇒ and / or (circle)

Step 3: $x \le$ _____ or $x \ge$ _____

Answer: _____

Graph it:

3. $|x-1|+2 < 6$

Isolate: $|x-1| <$ _____

Step 1 — Boundaries: $x-1 =$ _____ and $x-1 =$ _____ ⇒ $x =$ _____, $x =$ _____

Step 2 — Test $x = 1$: True / False In / Out ⇒ and / or

Step 3: _____$< x <$ _____

Answer: _____

Graph it:

4. $3|x+2| > 9$

Isolate: $|x+2| >$ _____

Step 1 — Boundaries: $x+2 =$ _____ and $x+2 =$ _____ ⇒ $x =$ _____, $x =$ _____

Step 2 — Test $x = -2$: True / False In / Out ⇒ and / or

Step 3: $x <$ _____ or $x >$ _____

Answer: _____

Graph it:

5. $|x+4|-1 \leq 3$

Isolate: $|x+4| \leq$ _____

Step 1 — Boundaries: $x+4=$ _____ and $x+4=$ _____ $\Rightarrow x=$ _____, $x=$ _____

Step 2 — Test $x=-4$: True / False In / Out $\Rightarrow$ and / or

Step 3: _____$\leq x \leq$ _____

Answer: _____

Graph it:

6. $4|x-3| \geq 8$

Isolate: $|x-3| \geq$ _____

Step 1 — Boundaries: $x-3=$ _____ and $x-3=$ _____ $\Rightarrow x=$ _____, $x=$ _____

Step 2 — Test $x=3$: True / False In / Out $\Rightarrow$ and / or

Step 3: $x \leq$ _____ or $x \geq$ _____

Answer: _____

Graph it:

Part D Challenge

Show all 3 steps on your own — no scaffold

1. $|2x-4| \leq 6$

Step 1 (boundaries): _____

Step 2 (test the middle, True/False, In/Out, and/or): _____

Step 3 (compound inequality, solved): _____

Answer: _____

Step 4 — Graph it:

2. $|3x+3| \geq 9$

Step 1 (boundaries): _____

Step 2 (test the middle, True/False, In/Out, and/or): _____

Step 3 (compound inequality, solved): _____

Answer: _____

Step 4 — Graph it:

3. $|2x + 1| < 5$

Step 1 (boundaries): _____

Step 2 (test the middle, True/False, In/Out, and/or): _____

Step 3 (compound inequality, solved): _____

Answer: _____

Step 4 — Graph it:

4. $|4x - 8| \geq 0$ *Hint: can an absolute value ever be negative?*

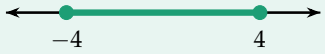
Answer: _____

Graph it:

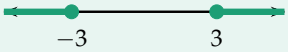
Answer Key (Teacher Copy)

Part A — each: boundaries, center test, and/or, final answer, graph

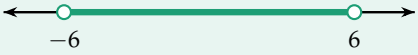
1. $x = -4, 4$; True; **In, and** $\Rightarrow -4 \le x \le 4$



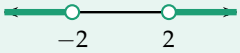
2. $x = -3, 3$; False; **Out, or** $\Rightarrow x \le -3$ **or** $x \ge 3$



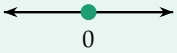
3. $x = -6, 6$; True; **In, and** $\Rightarrow -6 < x < 6$ (open circles)



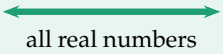
4. $x = -2, 2$; False; **Out, or** $\Rightarrow x < -2$ **or** $x > 2$ (open circles)



5. $x = 0$ **(single point)**

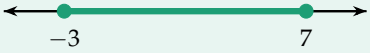


6. **All real numbers**

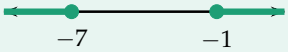


Part B

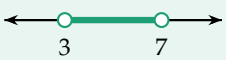
1. $x = -3, 7$; True; **In, and** $\Rightarrow -3 \le x \le 7$



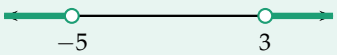
2. $x = -7, -1$; False; **Out, or** $\Rightarrow x \le -7$ **or** $x \ge -1$



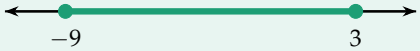
3. $x = 3, 7$; True; **In, and** $\Rightarrow 3 < x < 7$ (open circles)



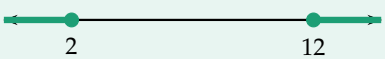
4. $x = -5, 3$; False; **Out, or** $\Rightarrow x < -5$ **or** $x > 3$ (open circles)



5. $x = -9, 3$; True; **In, and** $\Rightarrow -9 \le x \le 3$

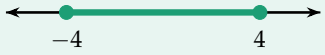


6. $x = 2, 12$; False; **Out, or** $\Rightarrow x \le 2$ **or** $x \ge 12$

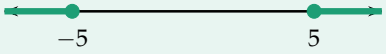


Part C

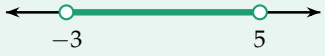
1. $|x| \leq 4$; $x = -4, 4$; True; **In, and** $\Rightarrow$ $-4 \leq x \leq 4$



2. $|x| \geq 5$; $x = -5, 5$; False; **Out, or** $\Rightarrow$ $x \leq -5$ **or** $x \geq 5$



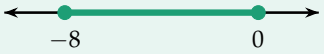
3. $|x - 1| < 4$; $x = -3, 5$; True; **In, and** $\Rightarrow$ $-3 < x < 5$ (open circles)



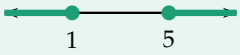
4. $|x + 2| > 3$; $x = -5, 1$; False; **Out, or** $\Rightarrow$ $x < -5$ **or** $x > 1$ (open circles)



5. $|x + 4| \leq 4$; $x = -8, 0$; True; **In, and** $\Rightarrow$ $-8 \leq x \leq 0$

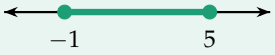


6. $|x - 3| \geq 2$; $x = 1, 5$; False; **Out, or** $\Rightarrow$ $x \leq 1$ **or** $x \geq 5$

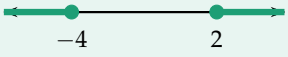


Part D

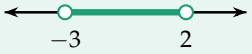
1. boundaries $-1, 5$; True; **In, and** $\Rightarrow$ $-1 \leq x \leq 5$



2. boundaries $-4, 2$; False; **Out, or** $\Rightarrow$ $x \leq -4$ **or** $x \geq 2$



3. boundaries $-3, 2$; True; **In, and** $\Rightarrow$ $-3 < x < 2$ (open circles)



4. **All real numbers**

